

TAKEOFF

☐ Fuel Pump — ON

☐ Fuel Tank - FULLEST TANK

☐ Lights — AS REQUIRED

☐ Flaps — SET 10° (25° SHORT/SOFT)

☐ Fuel/Oil Gauges - GREEN

☐ Airspeed Alive — CHECK

☐ Rotate —**65-70 MPH**

☐ Climb —**Vy 105 MPH / Vx 95 MPH**

✈️ IFR — DEPARTURE / ENROUTE

☐ Clearance — REVIEW (route / altitude / frequencies)

☐ Bugs — HDG / ALT / MINs SET

☐ Navigation Source — CONFIRM

☐ Instruments — CHECK (gyro / CDI sensing)

☐ Fuel Pump — ON

☐ Lights — AS REQUIRED

IFR — APPROACH CONFIGURATION

☐ Power — SET

☐ Pitch — SET

☐ Configuration — FLAPS 10°

☐ Airspeed — 110 MPH

☐ Sink Rate — 500 FPM

☐ Garmin 530W approach activation — LOAD / VERIFY

☐ DADS — CHECK every 1 NM or 500 FT

☐ 6 Ts — APPLY / REVIEW on change

Goal: Stabilized by FAF

IFR PERFORMANCE PROFILES

Phase	Power	Pitch	Config	Speed	V/S
Cruise	~15 in MP	0°	Clean	—	—
Descent	Low / Idle	−2 to −3°	Clean	138 MPH	−500 fpm
Approach Level 1	~15 in MP	0 to −1°	25°	104 MPH	−500
Approach Level 2	~17 in MP	0°	33°	104 MPH	−800
Missed	FULL	+7°	Clean	97 MPH	+800

V SPEEDS

☐ V1 - Abort Takeoff - 46 MPH (50% RWY)

☐ Vs0 - stall w/ flaps - 63 MPH (55kts)

☐ Vs1 - stall clean - 71 MPH (62kts)

☐ Vr - rotate - 65 MPH (56kts)

☐ Vy - best rate - 105 MPH (91kts)

☐ Vx - best angle - 95 MPH (83kts)

☐ BEST GLIDE - 100 MPH (87kts)

☐ Vfe - flaps extend - 125 MPH (109kts)

☐ Va - maneuvering - 149 MPH (129kts)

☐ Vno - max cruise - 168 MPH (146kts)

☐ Vne - never exceed - 212 MPH (184kts)

LANDING / GO-AROUND

☐ Downwind — 110 MPH

☐ Abeam - 90-95 MPH

☐ Over the #s - 80 MPH

☐ Flaps — AS REQUIRED (10° REC)

☐ Power — FULL

☐ Carb Heat — OFF

☐ Flaps — RETRACT IN STAGES

☐ Airspeed — ≥95 MPH

MISSED APPROACH

IMMEDIATE ACTIONS

☐ Power — FULL

☐ Pitch — +7°

☐ Configuration — CLEAN

☐ Airspeed — 97 MPH

☐ Climb — +800 fpm

☐ EMERGENCY & ABNORMAL OPERATIONS

CRITICAL MEMORY ITEMS

(Always accessible)

☐ Radio call — PAN-PAN / MAYDAY as appropriate on 121.5

☐ Squawk — 7700

☐ Engine Failure —**100 MPH**

☐ Engine Fire (In-Flight) —**Mixture IDLE CUTOFF**

☐ Electrical Fire / Smoke —**MASTER OFF**

☐ NORMAL OPERATIONS

☐ CLEAR RUNWAY

- ☐ Trim - TAKEOFF
- ☐ Flaps - UP
- ☐ Fuel Pump - OFF
- ☐ Landing Light - OFF
- ☐ Transponder - 1200

☐ SHUTDOWN

- ☐ Avionics - OFF
- ☐ Magnetos - CHECK
- ☐ Throttle - 1/4 INCH
- ☐ Mixture - FULL OUT/CUTOFF
- ☐ Magnetos - OFF
- ☐ Master - OFF
- ☐ Control Lock - SECURE
- ☐ Flight Plan - CLOSED

☐ PREFLIGHT

☐ ► FLOW

- ☐ Covers & Wheel Chalks - REMOVE
- ☐ Paperwork - ARROW-AV1ATE
- ☐ Control Lock - REMOVE
- ☐ Flaps - DOWN
- ☐ Trim - TAKEOFF
- ☐ Master - ON
- ☐ Lights - ON & CHECK
- ☐ Pitot Heat - ON & CHECK
- ☐ Stall Light - CHECK
- ☐ Fuel Quantity — CHECK / PROPER TANK
- ☐ Lights - OFF (KEEP BEACON ON)
- ☐ Pitot Heat - OFF
- ☐ Master - OFF
- ☐ Airframe & Hinges - WALK AROUND & CHECK
- ☐ Cowling - SECURE
- ☐ Tires & Brakes - CHECK
- ☐ Prop and Intake - CHECK
- ☐ Fuel Tanks - SUMP, MEASURE
- ☐ Oil — CHECK (6-10 QTS)
- ☐ Tow Bar - STOW
- ☐ Baggage - SECURE

☐ SAFETY CHECKLIST

- ☐ Seatbelts - EXPLAIN & ON
- ☐ Air vents/Heat - LOCATE & EXPLAIN
- ☐ First Aid & Fire Extinguisher - LOCATE
- ☐ Exits - EXPLAIN & LOCK
- ☐ Talking - ASSIST TRAFFIC
- ☐ Your Questions - ASK

☐ ENGINE START

- ☐ Brakes - SET
- ☐ Flaps - UP
- ☐ Electrical - ALL OFF
- ☐ Circuit Breaker - ALL IN
- ☐ Carburetor Heat - OFF
- ☐ Controls — FREE / CORRECT
- ☐ Prop - FULL FORWARD
- ☐ Mixture - RICH
- ☐ Fuel Selector - STARTUP SIDE
- ☐ Master - ON
- ☐ Throttle - 1/4 INCH OPEN
- ☐ Key/Mags - BOTH
- ☐ Area - CLEAR PROP
- ☐ Fuel Pump - ON
- ☐ Primer - 2 TO 3 OR AS REQUIRED
- ☐ Starter - CRANK ENGINE
- ☐ Throttle - 1000 RPM
- ☐ Avionics Master - ON
- ☐ Fuel Pump - OFF

☐ GROUND CHECK

- ☐ Doors & Windows - CLOSED
- ☐ Seatbelts - SECURE
- ☐ Controls - CHECK
- ☐ Trim - TAKEOFF
- ☐ Airspeed Indicator - READS 0
- ☐ Attitude Indicator - <5° BANK AND WINGS LEVEL
- ☐ Altimeter - FIELD ELEVATION
- ☐ VSI - Zero
- ☐ Turn & Slip Indicator - FLAGS OFF, WINGS LEVEL, FULL FLUID
- ☐ Heading Indicator - SET TO COMPASS HEADING
- ☐ Garmin 530W database — CHECK / UPDATE
- ☐ Garmin 530W instrument check — VERIFY

☐ RUN-UP

- ☐ Brake — SET
- ☐ Throttle — 2000 RPM
- ☐ Magnetos — CHECK (175 RPM MAX/50 RPM DIFFERENCE)
- ☐ Propeller — CHECK / FULL FORWARD
- ☐ Fuel/Oil Gauges — GREEN
- ☐ Ammeter - POSITIVE AMPS
- ☐ Suction — CHECK 5.0" +/- 1" HG
- ☐ Carburetor Heat — ON
- ☐ Throttle — IDLE / SMOOTH
- ☐ Carburetor Heat - OFF
- ☐ Throttle - 800-1000 RPM
- ☐ Flight Plan - ACTIVATE
- ☐ Stable RPM
- ☐ No abnormal vibration
- ☐ Engine indications normal

☐ IFR ACRONYMS / MNEMONICS

- ☐ IFR - Instrument Flight Rules
- ☐ VFR - Visual Flight Rules
- ☐ CDI - Course Deviation Indicator
- ☐ DME - Distance Measuring Equipment
- ☐ RNAV - Area Navigation
- ☐ ILS - Instrument Landing System
- ☐ FAF - Final Approach Fix
- ☐ MAP - Missed Approach Point
- ☐ MSA - Minimum Safe Altitude
- ☐ DA/MDA - Decision Altitude / Minimum Descent Altitude
- ☐ WAAS - Wide Area Augmentation System
- ☐ RAIM - Receiver Autonomous Integrity Monitoring
- ☐ ETE - Estimated Time Enroute
- ☐ ETA - Estimated Time of Arrival
- ☐ QNH - Altimeter setting to sea level
- ☐ DADS - Distance, Altitude, Direction, Speed. Check every 1 NM or 500 FT.
- ☐ 6 Ts - Turn, Time, Twist, Throttle, Talk, Track.
- ☐ GUMPF - Gas, Undercarriage, Mixture, Propeller, Flaps.
- ☐ GUMPS - Gas, Undercarriage, Mixture, Propeller, Seatbelts.
- ☐ CRAFT - Clearance, Route, Altitude, Frequency, Transponder.
- ☐ PAVE - Pilot, Aircraft, enVironment, External pressures.
- ☐ IMSAFE - Illness, Medication, Stress, Alcohol, Fatigue, Emotion.
- ☐ ARROW - Airworthiness certificate, Registration, Radio license, Operating handbook, Weight and balance.
- ☐ ATOMATOFLAMES - Common required instruments and equipment.
- ☐ 3 Ps - Perceive, Process, Perform.
- ☐ 5 Ps - Plane, Pilot, Passengers, Programming, Plan.

☐ ENGINE FAILURE — IN FLIGHT

- ☐ Radio call — MAYDAY, MAYDAY, MAYDAY on 121.5
- ☐ Squawk — 7700
- ☐ Airspeed — **100 MPH (BEST GLIDE)**
- ☐ Fuel Selector — FULLEST TANK
- ☐ Fuel Pump — ON
- ☐ Mixture — FULL RICH
- ☐ Carb Heat — ON
- ☐ Throttle — OPEN
- ☐ Power restored → Resume normal operations

☐ ENGINE RESTART — IN FLIGHT (POH)

- ☐ Fuel Selector — FULLEST
- ☐ Fuel Pump — ON
- ☐ Mixture — FULL RICH
- ☐ Carb Heat — ON
- ☐ Throttle — 1" OPEN
- ☐ Magnetos — BOTH
- ☐ Primer — AS REQUIRED
- ☐ Engine Gauges — CHECK

☐ FORCED LANDING

- ☐ Airspeed — **100 MPH**
- ☐ Landing Site — SELECT
- ☐ Mixture — IDLE CUTOFF
- ☐ Fuel Selector — OFF
- ☐ Fuel Pump — OFF
- ☐ Magnetos — OFF
- ☐ Master — OFF
- ☐ Seatbelts — SECURE
- ☐ Door — UNLATCH
- ☐ Flaps — AS REQUIRED

■ ENGINE FIRE — IN FLIGHT

- ☐ Radio call — MAYDAY, MAYDAY, MAYDAY on 121.5
- ☐ Squawk — 7700
- ☐ Mixture — IDLE CUTOFF
- ☐ Fuel Selector — OFF
- ☐ Fuel Pump — OFF
- ☐ Magnetos — OFF
- ☐ Cabin Heat / Air — OFF
- ☐ Airspeed **125 MPH (TOP WHITE ARC)** → Proceed to **Engine Failure / Forced Landing**

■ ✂ ELECTRICAL FIRE / SMOKE (POH-ALIGNED)

- ☐ Radio call — MAYDAY or PAN-PAN on 121.5
- ☐ Squawk — 7600 if comms fail; otherwise 7700
- ☐ Master Switch — OFF
- ☐ All Electrical Switches — OFF
- ☐ Cabin Heat — OFF
- ☐ Air / Vents — OPEN (ventilate)
- ☐ Smoke dissipating
- ☐ Master — ON
- ☐ Essential Circuits — **ONONE AT A TIME**
- ☐ If smoke returns → Master OFF → **Land ASAP**

■ ⚠ ALTERNATOR FAILURE / NO CHARGE

- ☐ Radio call — advise ATC / request assistance and vectors
- ☐ Squawk — as assigned; 7600 if radios fail
- ☐ Electrical Load — REDUCE
- ☐ Alternator Field CB — CHECK / RESET
- ☐ Alternator Output CB — CHECK / RESET
- ☐ Alternator — OFF (30 sec) → ON
- ☐ Ammeter shows system load
- ☐ If no output → Alternator OFF → Minimal load → **Land ASAP / Practical**